INTERNSHIP REPORT

Gen AI Virtual Internship

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Batch: 2026-27

B-Tech (CSE)

School of Computer Science & Engineering

IILM University, Greater Noida

Program Introduction

This report details the 8-week AICTE–EduSkills Virtual Internship Program focused on AWS Gen AI, completed from April to June 2026.

Designed by AWS Academy, the curriculum bridges the gap between theoretical knowledge and practical exposure to industry-relevant artificial intelligence technologies.

AWS Academy | Machine Learning | Generative AI

The Industry Challenge

Theoretical vs. Practical

While students often learn the theoretical foundations of AI and Machine Learning, they frequently lack hands-on, structured exposure to deploying these systems in real-world scenarios.

The Application Gap

Without practical experience, it is difficult to grasp how foundation models, prompt engineering, and Natural Language Processing are securely and responsibly applied to solve modern business problems.

Core Learning Objectives

Machine Learning

Master problem-solving using Amazon SageMaker, covering data splitting and model training workflows using the XGBoost algorithm.

NLP Applications

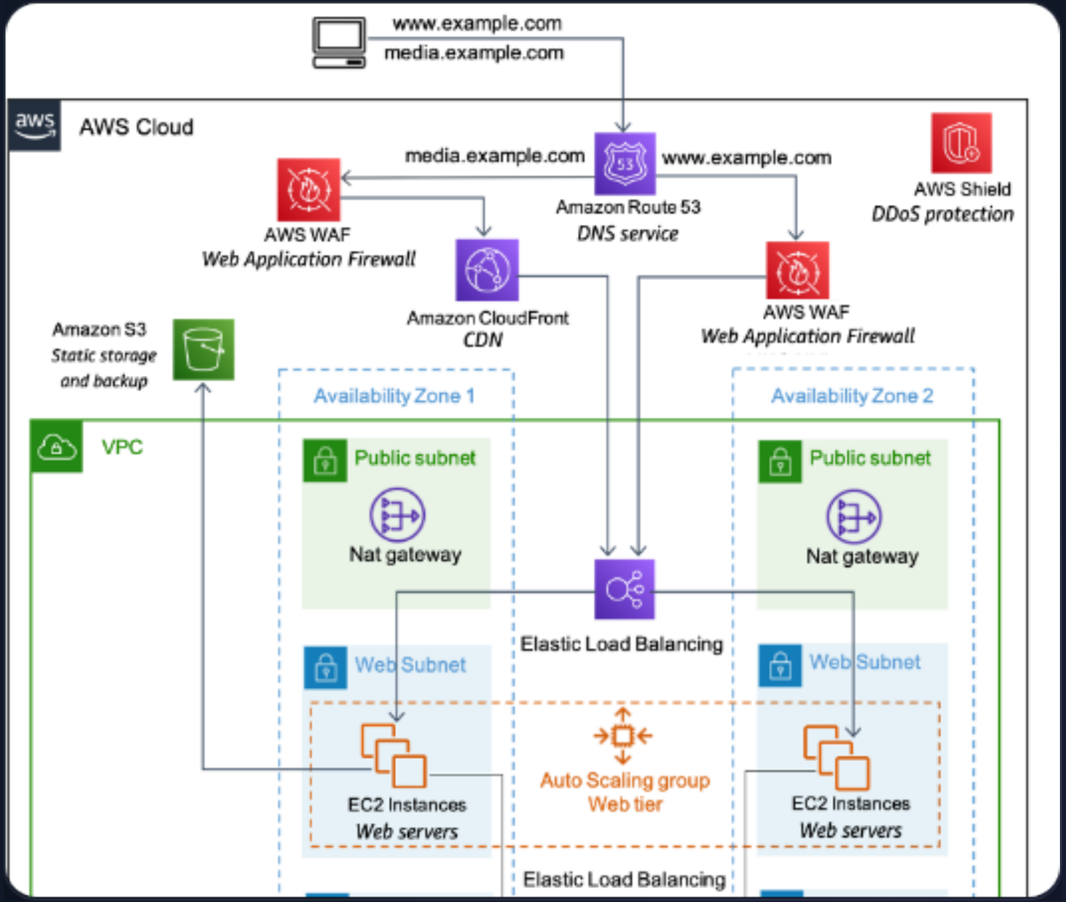
Understand text processing, sentiment analysis, information extraction, and topic modeling for real-world NLP applications.

Generative AI

Develop a working knowledge of Foundation Models, advanced prompt engineering, and the ethical, responsible deployment of AI.

Ecosystem & Platforms

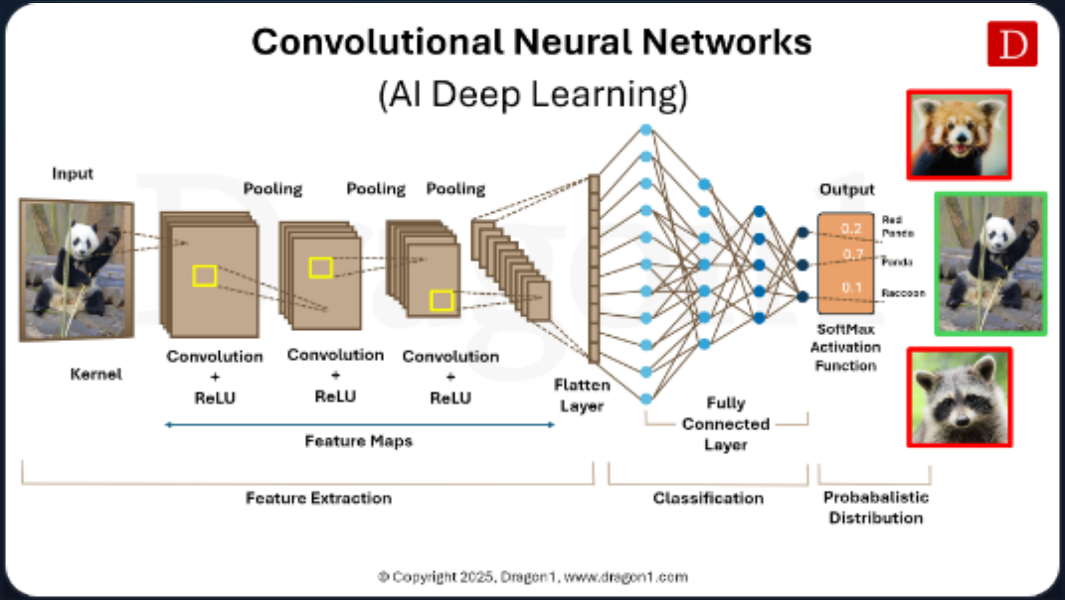
- **EduSkills Academy:** Operating under "Nation Building Through Skills," providing industry-oriented learning opportunities.
- **AICTE Portal:** Conducted under the prestigious AICTE National Internship Portal framework.
- **AWS Academy:** Provided the official, comprehensive curriculum covering cloud-native Gen AI implementations.



Technologies & Tools Deployed

- **Cloud AI:** Amazon SageMaker, XGBoost algorithm for model training.
- **Computer Vision:** AWS managed services for image and video analysis.
- **Language:** NLP, Sentiment Analysis, Topic Modeling.
- **Development:** Prompts, Prompt Engineering, Amazon CodeWhisperer.

System Architecture Layers



- **Machine Learning Layer:** SageMaker and XGBoost for core intelligence and forecasting.
- **NLP Layer:** Text processing and sentiment extraction engines.
- **Prompt Engineering:** Directing and refining interactions with Generative systems.
- **Responsible AI Layer:** Ensuring security, governance, and compliance bounds.

8-Week Program Methodology

Weeks 1 & 2

ML Foundations: SageMaker problem-solving, data splitting, XGBoost models, and forecasting.

Weeks 3 & 4

AWS & NLP: Amazon CodeWhisperer introduction and core Natural Language Processing fundamentals.

Weeks 5 & 6

Advanced NLP: Sentiment analysis, information extraction, topic modeling, and language processing.

Weeks 7 & 8

Generative AI: Foundation Models, prompt engineering, Responsible AI (security & governance), and Final Assessment.

Final Outcomes & Achievements

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OUTSTANDING GRADE

Demonstrated Competency

Successfully completed all weekly assessments, assignments, and project documentations.

Achieved comprehensive, practical knowledge spanning from baseline Machine Learning (SageMaker) to complex Generative AI Foundation Models and ethical AI frameworks.

Thank You!

Akee Kumar

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AICTE – EduSkills Virtual Internship Program

Image Sources

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